Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



TEAM DETAILS · AI BUILDER CUP 2026 · JAPAC · SUSTAINABILITY

Team name: Agent Ram

Team leader: Ramasamy P

Product: TerraSentinel AI — ground-motion early warning

Problem: After each Sentinel-1 pass, operators face thousands of InSAR points with no consequence ranking. TerraSentinel cuts motion into 250 m sites, ranks who inspects today (RPI), reports honest time-to-threshold bands (never failure dates), and uses Gemini to draft a number-verified inspection ticket.

TERRASENTINEL AI · STORY

Satellites see the ground move. Gemini tells engineers where to go first.

Each Sentinel-1 pass → ranked Act list → honest threshold band (never a failure date) → number-verified ticket.

1
Pass
Sentinel-1 every 6–12 d

2
Map
Points → 250 m sites

3
Rank
RPI × consequence

4
Explain
Gemini + number guard

5
Act
Crew · next pass

87 → 7
sites to Act list

RPI 86
NH-58 CRITICAL band

35–64 d
outlook if trend holds

GCP
Run · BigQuery · Gemini

BRIEF ABOUT THE IDEA

Ground creeps for months. Nobody ranks which creep matters.

Mountain highways · NH-58
Sole valley access. One slip cuts towns & supply.

Metro tunnels · Chennai
Settlement under roads & buildings. Tight tolerance.

Dams & tailings
Crest / toe motion with severe downstream consequence.

The decision we own
Which 250 m needs an engineer, why, and how soon?

Today's options fail the decision
- Raw InSAR portal — no asset link, no who-goes-today
- Rainfall alarm — trigger treated as hazard
- Generic AI — invents mm and failure dates
- Walk-over calendar — ignores acceleration now

We do
- Creep on linear assets
- Rank · explain · track
- Threshold window + assumptions

We refuse
- Failure / collapse dates
- Sudden inter-pass events
- Certified safety claim

OPPORTUNITIES · USP

Not an InSAR viewer — a decision layer with honest outlooks and guarded GenAI.

How different?

- 250 m sites on corridor geometry
- RPI = hazard x consequence x confidence
- Alerts = pass diffs; Act list = today
- Rain amplifies only (+50% cap)
- USGS quakes: context, never scored
- Gemini narrates; engine decides

How it solves it

- Validate → locate → measure
- WEAK guard: bad radar != CRITICAL
- Rank Act-today shortlist
- Outlook = ETA band, never a date
- Ticket: summary · cause · checklist
- Same maths: local XOR BigQuery

USP / demo value

- Honest continuing-regime bands
- Number guard rejects invented mm
- Production GCP posture (Run + BQ)
- 62,330 obs → 87 sites → 7 action
- 4 Act today in Chamoli
- Judge can sign in and act live

FEATURES

Every surface ships a decision — not a dashboard toy.

Command Center

Map + KPIs + Act today, ranked by consequence

Site drawer

RPI, speed, rain, outlook band, pass trend

Alerts

What changed between passes

Brief · Trends · Portfolio

Planning views; history recomputed from data

Work Orders + Gemini

Template-first; 7 tools; number guard

Data Hub

Validate-then-import; rain fetch; provenance; BQ sync

Access & audit

RBAC · sessions · CSRF · audit log

Methodology

Published formulae — every score auditable

STORY & PROCESS FLOW

One satellite pass — from space to a crew at the right 250 metres.

1 · Pass
Every 6–12 days

Sentinel-1 measures mm motion through cloud & night

2 · Map
Onto the corridor

1,355 points matched to highway / metro / dam → 87×250 m sites

3 · Rank
What matters

Fast + accelerating + critical + good data rises; rain alone never alarms

4 · Gemini
Write the ticket

7 tools → reason → JSON → number guard → field checklist

5 · Next
Keep watching

Crew acts; next pass shows calm-down or worse

Demo funnel (labelled synthetic)

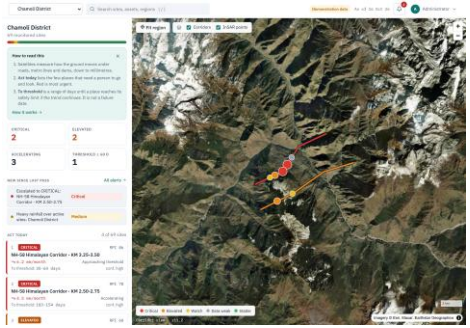
62,330	1,355	87	7	4
obs	pts	sites	action	Act today

Who it serves

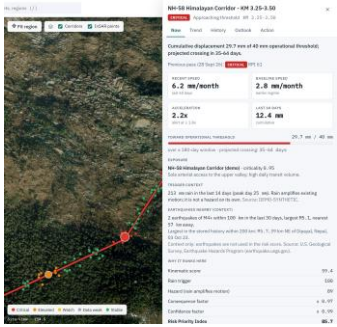
- Public: find problems while small — keep roads open
- Engineers: short list + ticket (what / where / why)
- Leaders: spend crews where risk is rising

PRODUCT UI (LIVE)

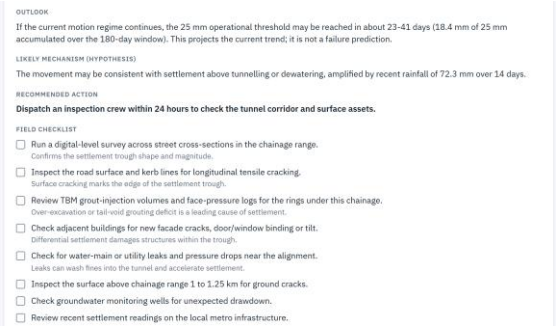
Running product screens — aspect-correct, not stretched wireframes.



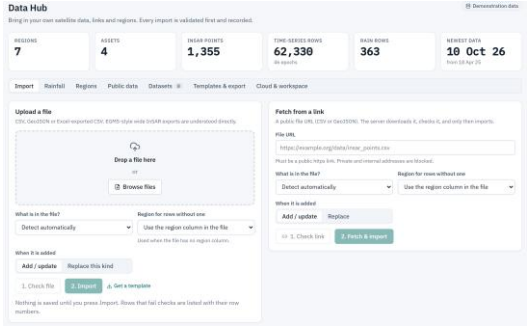
Command Center · Act list



Site · RPI & outlook



Gemini work order



Data Hub · validate then import

ARCHITECTURE STORY

Request flow on Google Cloud — one container, two storage backends, same maths.

Ingest	Store	Compute	Snapshot	Serve
Data Hub validates files & HTTPS links	BigQuery or local GIS join · history	Detection engine RPI · tier · band	Pass replay alerts · trends	UI + Gemini ticket audit trail

EDGE	Cloud Run · HTTPS · CSP/HSTS · rate limits · sessions · GCS state
UI + API	React/MapLibre · FastAPI · RBAC · CSRF · SSRF-safe fetch · audit
ENGINE	Deterministic RPI / tiers / ETA bands / alerts — GenAI never computes these
GENAI	Gemini tools → number guard → template fallback (Vertex or AI Studio)
DATA	BigQuery partitions/GEOGRAPHY XOR local · USGS context only · provenance

TECHNOLOGIES & GENAI FLOW

Google Cloud end-to-end. Gemini turns a verified score into an action.

Without GenAI an engineer gets “RPI 86 · CRITICAL”. With Gemini + our guard they get a ticket: what is happening, likely cause (hypothesis), what to do by when, and a site checklist — every number from tools.

Ingest
Cloud Run
Data Hub

Store
BigQuery
GIS + history

Compute
Engine
scores sites

Explain
Gemini
+ number guard

Act
Map · ticket
· audit

Gemini loop

- 7 read-only tools (metrics → quality)
- Round 0 must call tools → investigate → JSON
- Guard rejects invented mm / collapse language
- Fallback models; template always ships

GCP services

- Cloud Run — SPA + API + engine + agent
- BigQuery — partitions, spatial join, bytes cap
- Vertex AI / AI Studio — Gemini
- Secret Manager · GCS · ADC (no key files)

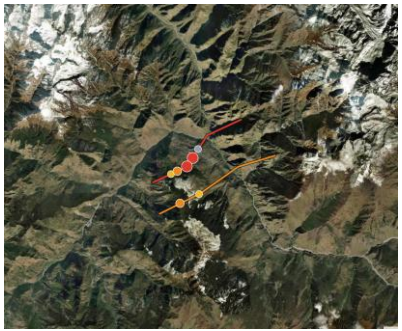
ESTIMATED COST (OPTIONAL)

GenAI only on tickets; ranking stays deterministic CPU.

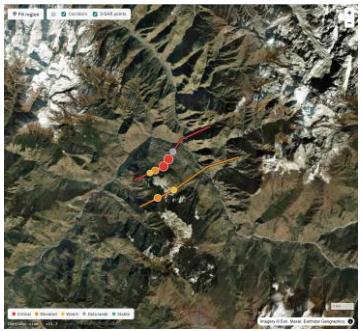
Cloud Run	1–2 vCPU / 2 GiB · min=1 while judging	low tens USD/mo warm
BigQuery	~62k rows · TTL cache · max bytes billed	pennies–low USD demos
Gemini Flash	Work-order path only · template fallback	per-ticket, optional
Secrets + GCS	Admin/Gemini secrets · state volume	pennies
Scale path	Materialize assignment · raise instances	linear with corridors

PROTOTYPE SNAPSHOTS

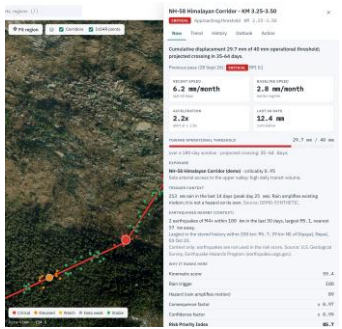
Demo value on screen — Chamoli, NH-58 CRITICAL, guarded ticket.



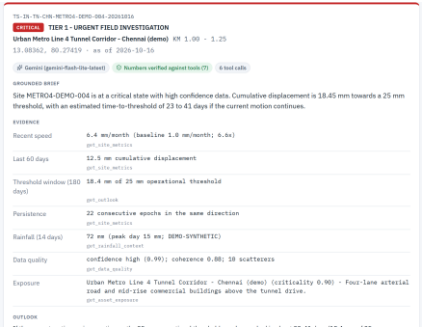
Chamoli · 69 sites · 4 act today



Ranked map — not every scatterer



NH-58 · RPI 86 · 35-64 day band



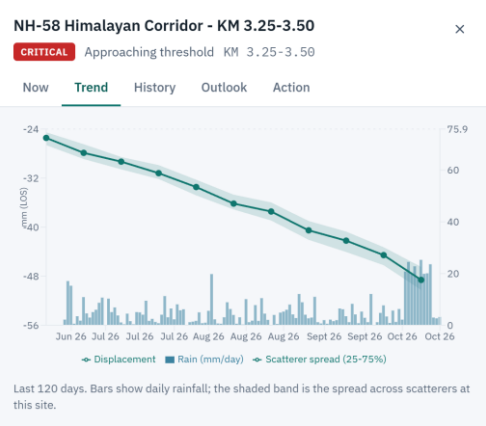
Gemini · numbers verified

PERFORMANCE / BENCHMARKING

NH-58 KM 3.25-3.50 — same numbers in UI, tools, and work order.

Worked example · CRITICAL · RPI 86

- Inputs: 6.2 vs 2.8 mm/mo · 2.2x · 29.7/40 mm · rain 213 mm
- kin = 100*(.4*0.62+.3*0.40+.3*0.74) = 59.4
- hazard = 59.4*(1+0.5*1.0) = 89.1 (rain capped)
- RPI = 89.1*0.975*0.985 ~ 85.7 → CRITICAL
- Outlook 35-64 days if trend continues — NOT a failure date
- History: site 61→86 · action sites 3→7 over 16 passes
- What-if: minor road 57 ELEV · weak radar ≤15 · rain-only ~1



Site trend · displacement + rain

Live Cloud Run + BigQuery (warm)

summary ~0.4s · sites ~0.3s · alerts ~2s

Hot-path prefetch before history warm

GenAI never computes RPI / tier / ETA

FUTURE DEVELOPMENT

Scale the data plane — honesty rules stay fixed.

Roadmap

- Production InSAR on the same Data Hub schema
- Materialize point-assignment for national scale
- HA once sessions leave single-process files
- Field checklist / offline crew packet
- OIDC / IAP for enterprise buyers
- Deeper rain/geology as amplifiers only

Buyer · ask · honesty

- Buyer: asset-integrity / geotech teams
- What they buy: attention — 1,355 pts → 7 sites
- Ask: live demo + 3-min video + repo
- Outlook bands are never collapse dates
- Gemini narrates tools only
- Synthetic demo data labelled in UI

SUBMISSION LINKS

GitHub · 3-minute demo · live product.

GitHub public repository

https://github.com/ramasamyp-aiengineer/terra-sentinel-ai

Demo video (3 minutes)

docs/pitch/TerraSentinel-AI-Demo-3min.mp4 (upload Drive/YouTube for portal)

Final product (live)

https://terrasentinel-598899050820.asia-southeast1.run.app

Sign-in credentials shown on the login page · Judge walk: docs/JUDGE.md

RISK ENGINE MATHS

Measure → Score → Project. Deterministic, tested, explainable.

1 · Measure

- Recent speed: slope last 60 d
- Baseline: 270 to 90 d ago
- Accel = |recent|/max(|base|,0.5)
- Persistence · seasonal recovery
- Site = median in 250 m bin
- conf mixes coh·epochs·pts·SNR
- Weak data can never be CRITICAL

2 · Score

- kin=100*(.4s+.3a+.3p)
- hazard=kin*(1+0.5*rain/200)
- RPI=hazard*(0.5+0.5*crit)
- *(0.5+0.5*conf)
- Slow/far x0.2 · seasonal x0.35
- WEAK cap 15 · quakes never scored
- CRITICAL / ELEVATED / WATCH

3 · Project

- ETA_const = remaining / speed
- ETA_accel = quadratic root
- band=[min,max] <= 365 d
- Highway 40 mm / 180 d demo
- Metro 25 mm · Dam 30 mm
- ALWAYS a band
- NEVER a failure date

Google Cloud H2S

TerraSentinel AI

Thank you — Team Agent Ram

Which 250 metres should an engineer inspect today?

The engine ranks them. Gemini writes the ticket. Every number is checked.

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Live demo + GitHub + 3-min video

Google Cloud AI Builder Cup 2026 · Sustainability

Gemini · Vertex AI · BigQuery · Cloud Run · Sentinel-1